

Antioxidants as antidepressants: fact or fiction?

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Depression is a medical condition with a complex biological pattern of aetiology, involving genetic and epigenetic factors, along with different environmental stressors. Recent evidence suggests that oxidative stress processes might play a relevant role in the pathogenic mechanism(s) underlying many major psychiatric disorders, including depression. Reactive oxygen and nitrogen species have been shown to modulate levels and activity of noradrenaline (norepinephrine), serotonin, dopamine and glutamate, the principal neurotransmitters involved in the neurobiology of depression. Major depression has been associated with lowered concentrations of several endogenous antioxidant compounds, such as vitamin E, zinc and coenzyme Q10, or enzymes, such as glutathione peroxidase, and with an impairment of the total antioxidant status. These observations introduce new potential targets for the development of therapeutic interventions based on antioxidant compounds. The present review focuses on the possible role of oxidative stress processes in the pathogenesis of depression. The therapeutic potential of antioxidant compounds as a co-adjuvant treatment to conventional antidepressants is discussed. For instance, N-acetyl-cysteine has been shown to have a significant benefit on depressive symptoms in a randomized placebo-controlled trial. Additionally, curcumin, the yellow pigment of curry, has been shown to strongly interfere with neuronal redox homeostasis in the CNS and to possess antidepressant activity in various animal models of depression, also thanks to its ability to inhibit monoamine oxidases. There is an urgent need to develop better tolerated and more effective treatments for depressive disorders and several antioxidant treatments appear promising and deserve further study.